

USB-A Manual Switch Ver 2.0

A Physical USB Switch

Project Report | 2026 | Haoyang Jiang

Project Report Update — August 30, 2026:

Following the successful validation of the initial prototype (Ver 1.0) this revision—USB-A Manual Switch Ver 2.0—focuses on enhancing circuit protection, optimizing signal integrity for high-speed differential pairs, improving thermal/grounding performance, and streamlining mechanical assembly. While Ver 1.0 established the fundamental physical disconnection logic, Ver 2.0 upgraded the board to an industrial-grade, ESD-protected, single-sided SMT assembly design.

New Features:

Elements	Before	After	Reason
Circuit protection	no protection	Integrated SRV05-4 TVS Diode Array	USB interfaces are prone to ESD spikes during hot-plugging. The SRV05-4 provides sub-nanosecond clamping protection on VBUS, D+, and D- lines, making the standalone capacitor redundant while significantly boosting transient protection.
Signal Integrity (D+/D-)	Uncompensated differential traces passing through switch cutouts	Equalized \$D+/D-\$ traces using KiCAD single-track length tuning (meanders).	Because D+ routes through the DPDT switch while D- routes directly, an imbalance

			in trace length was introduced. Snake-style length tuning ensures identical electrical length, eliminating phase skew in USB 2.0 differential signaling.
Component Placement	Dual-side placement: USB ports & LEDs on Top (F.Cu); Switch on Bottom (B.Cu)	100% Single-sided Top Placement (all SMT/THT components on F.Cu).	Ver 1.0 flipped the switch to the bottom to avoid height interference, but this increased assembly complexity and required dual-sided SMT reflow/soldering. Adjusting component spacing allowed everything to fit cleanly on the top side for an aesthetically unified look and lower fabrication cost.
Grounding Strategy	Thin, discrete 0.5mm GND routing tracks.	Solid Ground Copper Pour (Top & Bottom Ground Planes with stitched vias).	Discontinuous ground tracks increase return path impedance and EMI radiation. A solid ground pour lowers ground loop impedance, enhances heat dissipation, and improves ESD immunity.
Components Adjustment	1000ohm current-limiting resistors	220 ohm	Adjusted resistor values to optimize LED brightness for clear visual indication under bright ambient lighting while remaining within safe current limits

New Features Overview:

